MD. Mehedi Hasan Maruf

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Summary

Software engineer with 3+ years of experience building backends, web apps, and AI features in TypeScript and Python. At Aamira Software Solutions I lead a small team and build TypeScript microservices and a retrieval-augmented generation (RAG) feature. Before that, at Code Studio, I wrote Django and Node backends and looked after the production Linux servers they ran on. I also do applied ML research: my B.Sc. thesis at Rajshahi University of Engineering & Technology (RUET) built a system that finds the university courses holding students back and checks that answer with causal inference. A paper from it is under review at LAK 2027. On the side I am building Omnious, a visual debugger that maps runtime errors onto a program's code graph.

Research

TRACE: Curriculum Bottleneck Detection & Causal Validation *Jul 2025 – Aug 2026*

B.Sc. Thesis, Dept. of CSE, RUET

- The question came from my own class. About a third of us failed one first-year mathematics course, while about 4% fail a typical course. Did that course hold students back, or only show who was already behind?
- Built TRACE to find courses like it. It ranks each course by its place in the prerequisite graph (NetworkX, GAT), a Bayesian model of its fail rate, and the tone of student feedback (DeBERTa, VADER).
- Tested how best to combine its signals on simulated curricula with planted bottlenecks (seven conditions, 30 curricula each). When each bottleneck showed up in only one signal, merging the signals' shortlists reached a top-8 precision of 0.65, against 0.32 for averaging their ranks. When bottlenecks showed up in every signal, averaging did better (0.60 against 0.41).
- Used causal inference to test what drives failure in a flagged module. On 23,369 Open University students (OULAD), a DoWhy backdoor-adjusted model estimated that high assignment engagement raised held-out pass rates by 19.9 percentage points ($p < 0.01$, E-value 1.89).
- After peer review, rebuilt the study into a paper, *When Agreement Misleads: Auditing Course Shortlists in Curriculum Analytics*, now under review at LAK 2027. It gives an exact bound on how much a combined shortlist can change top-k precision. In simulation, sources with a shared bias picked mostly the same courses (88% overlap), yet only 15% of those picks were real bottlenecks.

Experience

Software Engineer II *Jan 2025 – Present*

Aamira Software Solutions, Kokomo, Indiana, USA (remote)

- Lead three junior engineers on the company's React applications, after starting on its Angular project.
- Raised client response rates by 30% by leading development of a CRM system.
- Cut API response time by 40% by designing and building a TypeScript microservice backend.
- Made the application 60% faster by optimizing its Supabase (PostgreSQL) queries.
- Building a retrieval-augmented generation (RAG) feature for the platform.

Junior Software Engineer *Nov 2022 – Apr 2024*

Code Studio, Rajshahi, Bangladesh

- Kept 12 Linux servers at 99.8% uptime, and cut deployment time by 40% by reworking the CI/CD pipelines.
- Moved 5 client environments onto Docker, which made them simpler to deploy.
- Wrote JavaScript, TypeScript, and Python (Django) backends used by thousands of people a day.

AI Code Reviewer / RLHF Evaluator *Oct 2024 – Jan 2026*

Outlier (remote)

- Reviewed and fixed C++ and Python code for AI systems with 50,000+ daily users; helped write evaluation guidelines.
- Rated and annotated LLM responses (RLHF) in English and Bangla for correctness, consistency, and safety.
- On later projects, wrote and tested prompts, and labeled objects in images with written descriptions for vision models.

Industrial Trainee *Mar 2025*

Brain Station 23, Dhaka, Bangladesh

- Represented a 15-person trainee cohort to management.
- Built and deployed StudentAI (see Projects), an AI learning assistant on FastAPI and Next.js.

Competitive Programming Mentor *Aug 2023 – Nov 2023*

RAPL (University Organization)

- Taught competitive programming to RUET students, covering number theory and graph traversal algorithms.
- Organized biweekly coding contests.

Selected Projects

Omnious | Visual Debugging Platform *Nov 2025 – Present*

- Building a debugger that parses a TypeScript or JavaScript codebase into a graph of files, functions, and classes, then animates runtime traces over it and shows where errors cluster.
- An AI assistant suggests root causes tied to specific graph nodes, using embeddings in pgvector and a locally run LLM (Ollama). Community detection groups the graph into architectural layers.
- A TypeScript Turborepo monorepo: Fastify 5 and tRPC v11 backend, Next.js 16 / React 19 frontend, and PostgreSQL 15 on Supabase (pgvector, ltree, pg_trgm, row-level security).
- Ships a CLI indexer and Express and Next.js SDKs; runs in Docker Compose and deploys through GitHub Actions.

StudentAI | Intelligent Learning Companion *Mar 2025*

- An AI learning assistant (FastAPI, Next.js) built during industrial training at Brain Station 23.
- Answers questions about uploaded documents through a LangChain RAG pipeline with semantic retrieval and conversation memory; it also analyzes YouTube lectures and writes quizzes.
- Accepts PDFs, images, and video, and has real-time WebSocket chat, JWT login, and progress tracking.
- Deployed on Vercel and Render with a Supabase database.

PharmaC | Desktop Pharmacy Management System *Feb 2025*

- Built a WPF desktop app (.NET 8, MVVM, WPF-UI) for pharmacy inventory, sales, and orders.
- Modeled medicines, brands, generics, and admin accounts in Entity Framework Core (SQL Server, MySQL).
- Implemented JWT authentication with token refresh, barcode lookup for medicine scanning, and per-device licensing from a CPU/motherboard/BIOS hardware fingerprint.
- Connected the app to a REST API (`HttpClient`) for medicines, orders, carts, and licenses, so inventory syncs in real time and checkout runs in BDT.

Thesis Site | Django Web App *Mar 2024 – May 2024*

- Built a Django web app where students post thesis topics and talk with supervisors and the department.
- Built a responsive Bootstrap interface with full CRUD operations for students, supervisors, and administrators.

Skills

AI/ML: LLM applications, RAG, LangChain, Hugging Face Transformers (DeBERTa), PyTorch, GNNs (GAT), NLP, causal inference (DoWhy), Bayesian modeling, vector search (pgvector), model evaluation (RLHF)
Languages: TypeScript, Python, JavaScript, C++
Backend: NestJS, FastAPI, Fastify, tRPC, Django
Frontend: Next.js (React), Angular
.NET: .NET 8, WPF, Entity Framework Core, MVVM
Databases: PostgreSQL (Supabase), MongoDB, MySQL, SQL Server
DevOps: Docker, GitHub Actions (CI/CD), Linux server administration

Education

B.Sc. Engineering, Computer Science and Engineering *Mar 2022 – Aug 2026*

Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh

- CGPA: 3.75 / 4.00, Honours
- Thesis: *TRACE*, which ranks the courses that hold students back using prerequisite graphs, Bayesian fail-rate models, and student-feedback sentiment, then checks the result with causal inference on 23,369 student records. A follow-up paper is under review at LAK 2027.
- Relevant coursework: Machine Learning, Artificial Intelligence, Data Mining, Applied Statistics and Queuing Theory, Security and Privacy, Database Systems, Algorithm Analysis and Design, Theory of Computation.

Certifications & Achievements

- Digital Accessibility Training (DAT) — a2i (Aspire to Innovate), Govt. of Bangladesh (Oct 2025)
- Industrial Training Certificate — Brain Station 23 (Mar 2025)
- PHP (Laravel) — EDGE Project (Feb 2025)
- Experience Certificate — Code Studio (Apr 2024)
- 4th Place — RUET CodeSmash, team competitive programming contest (Mar 2023)
- 4th Place — RUET CodeSpark, individual competitive programming contest (Oct 2022)